

Q3 Campaign Targeting: Shift to Model-Guided Outreach

TO Marketing Director & CFO, NovaBank
FROM Ettore Bertolini
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RE Predictive Targeting Framework: Q3 Campaign Decision, Rationale, and Risks

1 DECISION

We recommend replacing indiscriminate outreach with a machine learning scoring model that ranks all 41,000 clients by their predicted likelihood to subscribe to a term deposit. For Q3, **contact only clients whose model score meets or exceeds a fixed probability threshold of 0.20**, which corresponds to approximately the top 20% of the ranked list (~8,200 clients). In production, the bank must use the fixed probability cutoff (≥ 0.20), not a percentile rank: a percentile always selects 20% of clients regardless of how scores shift, which breaks down under changing macroeconomic conditions (see Section 4). The remaining 80% of clients should not be contacted.

2 RATIONALE

NovaBank’s current approach contacts every client regardless of interest, yielding an 11% conversion rate and, on a direct profit basis, a net campaign loss of €7,000 per cycle. Cost assumptions are €20 per agent call and €150 expected margin per deposit signed. These figures are conservative estimates to be confirmed by the CFO before go-live; the business case remains positive down to a €50 margin. If leadership also values blanket outreach for brand awareness or relationship maintenance, those benefits can be modelled separately. Nevertheless, on pure campaign economics, indiscriminate calling is a net cost.

The model is a Gradient Boosting classifier trained on 33,000 historical client records to predict subscription likelihood before the call is placed. It was validated against a logistic regression baseline: the baseline achieved ROC-AUC of 0.80 and Average Precision of 0.46, while Gradient Boosting improved both to 0.81 and 0.49 respectively. Five-fold cross-validation confirmed stability (AP 0.459 ± 0.012). The model’s three strongest signals are macroeconomic conditions (employment level and Euribor rate, jointly accounting for over 50% of predictive weight) and each client’s prior engagement history. These are transparent, auditable drivers that compliance officers can review without concern. The model was tested for disparate impact across age and income segments; no statistically significant bias was detected. Age appears as a secondary factor (6% weight) used as a life-stage proxy, not as a prohibited characteristic. Its inclusion should be confirmed with compliance before launch (GDPR Article 22).

3 EXPECTED IMPACT

Metric	Blanket Outreach	Model-Guided (Top 20%)
Clients contacted	41,188	~1,648 (test set)
Conversions captured	100%	66% of all subscribers
Wasted calls (false pos.)	36,548	1,038
Gross revenue	€138,600	€91,500
Cost of wasted calls	−€730,960	−€20,760
Net operating profit	−€7,000	+€70,740 (ROI 4.4×)

Results in the table above reflect the held-out test set (20% of the full 41,188-client portfolio). Scaled to the full campaign pool, the model is projected to deliver approximately **€350,000–370,000 in net operating profit per cycle**, assuming stable conversion rates. The 34% of subscribers not captured represents €47,700 in forgone margin on the test set, a deliberate trade-off accepted because the precision gains of targeted outreach more than offset this loss. Clients the model flags with the highest confidence (score ≥ 0.60) convert at nearly five times the population average; sales teams should prioritise highest-score clients first.

4 KEY RISKS

HIGH Macro sensitivity. The model’s strongest predictor is employment level. Scenario testing shows a combined stress shock (Euribor +1%, employment −5%) reduces net profit from €70,740 to €2,890; an employment crisis alone (−5%) collapses the percentile-based rule to €130 by flagging 7,737 clients instead of 1,648. *Mitigation: enforce the fixed probability cutoff (≥ 0.20) as the production rule and re-score clients monthly as macro conditions evolve.*

MED Margin assumption. The business case rests on a €150 expected margin per deposit. If actual margin falls below €50, the campaign becomes marginally profitable at any threshold. *Mitigation: CFO to confirm margin estimate before Q3 go-live.*

MED Threshold decay. The ≥ 0.20 cutoff was calibrated on historical data. As client behaviour and economic

conditions shift, this threshold may flag too many or too few clients. *Mitigation: revalidate against a holdout set each quarter.*

Recommended next step: Run a 4-week pilot on 2,000 clients (50% model-selected, 50% random control) ahead of the full Q3 launch. Go/no-go criterion: model group conversion rate $\geq 2\times$ control rate *and* net profit $\geq$ €15,000 above control. Decision at week 5.

Full methodology, baseline vs. model comparison, scenario analysis, feature diagnostics, and AI usage log: [Slide Deck](#) | [Reproducible Notebook](#)